

Module Title: Stochastic Calculus for Finance

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School of Mathematics

Semester Two 2022/23

Calculator instructions:

- You are allowed to use a non-programmable calculator in this examination.

Dictionary instructions:

- You are not allowed to use your own dictionary in this exam. A basic English dictionary is available to use. Raise your hand and ask an invigilator if you need it.

Exam information:

- There are 4 pages to this examination.
- There will be **2 hours and 30 minutes** to complete this examination.
- This examination is worth 80% of the module mark.
- There are 3 questions in this exam paper. You must answer all 3 questions.
- Questions 1 and 3 count for 30 marks while Question 2 counts for 40 marks.
- The numbers in brackets indicate the marks available for each sub-question.
- You must show all your calculations.
- You must write your answers in the answer booklet provided. If you require an additional answer booklet, raise your hand so an invigilator can provide one.
- You must clearly state your name and Student ID Number in the relevant boxes on your answer booklet. Other boxes may be left blank.

1. (a) [8 marks] Let Ω be a set. Give the definition of a σ -algebra $\mathcal{F}$ on Ω .
- (b) [6 marks] Let $\Omega = \mathbb{R}$. Determine whether the following classes of sets are σ -algebras on Ω (simply state YES/NO):
- i. $\mathcal{F}_1 = \{\emptyset, \mathbb{R}\}$
 - ii. $\mathcal{F}_2 = \{\emptyset, \mathbb{R}, (0, 1), (-\infty, 0]\}$
 - iii. $\mathcal{F}_3 = \{\emptyset, \mathbb{R}, (-\infty, 0), [0, +\infty)\}$
- (c) [8 marks] Let $\Omega = [0, 1]$, let $\mathcal{F} = \mathcal{B}([0, 1])$ be the Borel σ -algebra on $[0, 1]$ and let $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ be the Lebesgue measure, that is, a probability measure with the property

$$\mathbb{P}((a, b]) = b - a, \quad \text{for all } a, b \in [0, 1] \text{ with } a < b.$$

Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable defined by

$$X(\omega) = \begin{cases} 5, & \text{if } \omega \in (0, 1/2] \\ 3, & \text{if } \omega \in (1/2, 2/3] \\ 2, & \text{if } \omega \in (2/3, 5/6] \\ 0 & \text{otherwise.} \end{cases}$$

- i. Calculate $\mathbb{E}[X]$.
- ii. Let Y be a non-negative random variable such that $\mathbb{E}[Y] = \mathbb{E}[X]$. Show that for any constant $K > 0$, we have

$$\mathbb{P}(Y > K) \leq \frac{10}{3K}.$$

- (d) [8 marks] Suppose that we flip infinitely many coins (countably many). For $n \in \mathbb{N}$, let $X_n = 1$ if the n -th coin is “Heads” and $X_n = 0$ if it is “Tails”. Suppose that the coins are not fair but we know that

$$P(X_n = 0) = \frac{n^2 - 1}{n^2}.$$

Show that with probability one, the sequence $(X_1, X_2, X_3, \dots)$ contains only finitely many 1s.

2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $T > 0$, and consider a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ consisting of σ -algebras $\mathcal{F}_t \subset \mathcal{F}$.
- (a) [5 marks] Give the definition of a one dimensional Wiener process $(W(t))_{t \in [0, T]}$ on $(\Omega, \mathcal{F}, \mathbb{P})$. When do we say that a Wiener process $(W(t))_{t \in [0, T]}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ is an $\mathbb{F}$ -Wiener process on $(\Omega, \mathcal{F}, \mathbb{P})$?
- (b) [6 marks] If $(W(t))_{t \in [0, T]}$ is a Wiener process and $T = 10$, calculate the expectation of the following random variables:
- $$(W(9))^2, \quad (W(9) - W(3))W(2), \quad W(6)W(2).$$
- (c) [5 marks] Let $(M(t))_{t \in [0, T]}$ be a stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$. When do we say that $(M(t))_{t \in [0, T]}$ is an $\mathbb{F}$ -martingale?

- (d) [5 marks] Suppose that $T = 1$. Let $(W(t))_{t \in [0, T]}$ be a Wiener process on $(\Omega, \mathcal{F}, \mathbb{P})$, and consider a process $(Y(t))_{t \in [0, T]}$ given by

$$Y(t) = X_1 \mathbf{1}_{(0, 1/2]}(t) + X_2 \mathbf{1}_{(1/2, 1]}(t),$$

where X_1 and X_2 are $\mathcal{F}_0$ -measurable and $\mathcal{F}_{1/2}$ -measurable, respectively, bounded random variables. Give the definition of the random variable

$$\int_0^1 Y(s) dW(s).$$

- (e) [9 marks] Let $(W(t))_{t \in [0, T]}$ be a Wiener process on $(\Omega, \mathcal{F}, \mathbb{P})$ and let $a, b \in \mathbb{R}$. Consider the Itô process

$$X(t) = at + bW(t).$$

Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that its first and second order derivatives u' and u'' exist and are continuous. Suppose that u' is bounded and that for all $x \in \mathbb{R}$ we have that

$$\frac{b^2}{2} u''(x) + au'(x) = 0.$$

Use Itô's formula to show that $u(X_t)$ is an $\mathbb{F}$ -martingale.

- (f) [10 marks] Let $(W(t))_{t \in [0, T]}$ be a Wiener process on $(\Omega, \mathcal{F}, \mathbb{P})$ and for $k \in \mathbb{N}$, set

$$m_k(t) := \mathbb{E}[(W(t))^k].$$

Use Itô's formula to show that for all $k \geq 2$, $t \in [0, T]$, we have

$$m_k(t) = \frac{1}{2} k(k-1) \int_0^t m_{k-2}(s) ds,$$

Moreover, find $\mathbb{E}[(W(t))^4]$ and $\mathbb{E}[(W(t))^6]$, for $t \in [0, T]$. You may use the fact that for each $k \in \mathbb{N}$, there is a constant C , such that for all $t \in [0, T]$, we have $\mathbb{E}[|W(t)|^k] \leq C$.

3. Let $T > 0$ and let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \in [0, T]}$ satisfying the usual conditions. Let $(W(t))_{t \in [0, T]}$ be a one-dimensional $\mathbb{F}$ -Wiener process on $(\Omega, \mathcal{F}, \mathbb{P})$. Consider the Black-Scholes model where there exists exactly one risky asset whose dynamics is given by

$$dS(t) = 2S(t) dt + \frac{1}{2} S(t) dW_t, \quad S(0) = 1, \quad t \in [0, T],$$

where $T > 0$ is the terminal time and the risk-free interest rate is given by $r = 0.05$.

- (a) [6 marks] Let $\mathbb{P}^*$ be the risk-neutral probability measure of the above model and recall that under $\mathbb{P}^*$ the process $W^*(t) = 2(2 - 0.05)t + W(t)$ is an $\mathbb{F}$ -Wiener process. Write down the stochastic equation that $(S(t))_{t \in [0, T]}$ satisfies under the measure $\mathbb{P}^*$ (driven by W^*) and write down a solution of this equation.

- (b) [10 marks] Use Itô's formula to show that the process that you gave as a solution is indeed a solution.
- (c) [5 marks] Are there any other solutions to this equation starting from $S(0) = 1$? If "yes", then provide another one, if "no" explain why.
- (d) [9 marks] Consider a derivative written on the risky asset which pays at the terminal time $T = 10$, the amount $\sqrt{S(10)}$ to the owner. Find the price of this derivative at time zero (give a real number and show your workings).

You can use that if $\theta \in \mathbb{R}$ and $X \sim N(0, 1)$, then the expectation of $\exp(\theta X)$ is given by $\exp(\frac{\theta^2}{2})$.